

FARHAD ZAKER

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DISSERTATION

Developing Thermogalvanics in Cementitious Composites for Simultaneous Structural and Heat-Harvesting Capabilities

Chair: Dr. Matthew Powell-Palm. **Committee:** Dr. Jaime Grunlan, Dr. Congrui Jin, Dr. Patrick Shamberger.

Status: Preliminary examination passed March 2026.

Establishes a novel framework for converting load-bearing Portland cement into a thermogalvanic waste heat harvesting cell, retaining compressive strength (≈ 30 MPa at 28 days) while delivering a thermopower of 5.4 mV K^{-1} and a peak power output of 6 W m^{-2} across a 40°C temperature gradient.

EDUCATION

Texas A&M University, College Station, Texas

Expected May 2027

Ph.D. in Mechanical Engineering (GPA: 3.68)

Sharif University of Technology, Tehran, Iran

Jan 2020

M.S. in Structural Engineering

Thesis: Online Safety Analysis Algorithm for Post-Disaster Inspections Using Image Processing.

Iran University of Science and Technology, Tehran, Iran

Oct 2016

B.S. in Structural Engineering

Thesis: Dispersion of Nanoparticles In Cementitious Matrices: Nanoparticle Geometry Perspective.

AWARDS & FELLOWSHIPS

Elizabeth & Raymond Marlow '53 Scholarship — Texas A&M University

2025

Full Graduate Research Assistantship — Texas A&M University

2022–2025

Summer Research Grant — Texas A&M University

2022–2024

PATENTS

2 provisional patents (May 2026)

1. **Zakertabrizi, M.**, Hosseini, E., Powell-Palm, M.J. “*Thermogalvanic Cementitious Composites*,” 2026. U.S. Provisional Application Pending.
2. Hosseini, E., **Zakertabrizi, M.**, Powell-Palm, M.J. “*Redox-Split Thermogalvanic Cells, Compositions Therefor, And Methods Of Making And Use Thereof*,” 2026. U.S. Provisional Application No. 63/962,992.

PUBLICATIONS

25 peer-reviewed publications • 1,441 citations • h-index: 18 • i10-index: 20 (Google Scholar, May 2026)

Manuscripts in Preparation

1. **Zakertabrizi, M.** et al. A Buffered Hexacyanoferrate Electrolyte for Thermogalvanic Heat Harvesting across Symmetric and Asymmetric Electrode Configurations. Manuscript under review. **[1st author]**
2. **Zakertabrizi, M.** et al. Thermogalvanic Cement for Low-Grade Thermal Energy Harvesting. Manuscript in preparation; target submission Summer 2026. **[1st author]**

Peer-Reviewed Journal Articles

1. Hosseini, E., **Zakertabrizi, M.**, Hosseini, M., Zarriz, A., Azhdari, R., et al. Thermally-driven chemical heterogeneity produces large thermopower and multiday operability in a thermogalvanic cell. *Journal of Materials Chemistry A* 14, 7695–7705 (2026). **[2nd author]**

2. **Zakertabrizi, M.**, Bozorgmehrian, F., Jeong, M., Hosseini, E., Ponce, V., et al. Patterned Nanostructures on Cathodes: A Pathway to Stronger, High-Energy, High-Power Li-Ion Batteries. *ACS Nano* 19(45), 38970–38980 (2025). [1st author]
3. Hosseini, E., **Zakertabrizi, M.**, Hosseini, M., Powell-Palm, M. J. On a Continuous Aqueous Thermogalvanic Redox Agent with Anomalous Thermopower. *Nano Letters* 25(31), 11986–11992 (2025). [2nd author]
4. **Zakertabrizi, M.**, Hosseini, E., Fallahi, H., Creasy, T., Tabei, A., Razmjou, A., et al. Dissecting atomic interweaving friction reveals the orbital overlap repulsion and its role in the integrity of woven nanofabrics in composites. *Advanced Composites and Hybrid Materials* 7(3), 86 (2024). [1st author]
5. Jangale, P., Hosseini, E., **Zakertabrizi, M.**, Jarrahashi, D. Surface tension and evaporation behavior of liquid fuel droplets at transcritical conditions. *Fuel* 358, 130187 (2024). [3rd author]
6. Kaynan, O., Hosseini, E., **Zakertabrizi, M.**, Motta De Castro, E., Pérez, L. M., et al. Multifunctionality through Embedding Patterned Nanostructures in High-Performance Composites. *Advanced Materials* 35(32), 2300948 (2023). [3rd author]
7. Shariatnia, S., **Zakertabrizi, M.**, Hosseini, E., Song, K., Jarrahashi, D., Asadi, A. Engineering Multimaterial Nanostructured Deposits by the Amphiphilicity Degree and Intermolecular Forces. *Advanced Materials Technologies* 8(7), 2201569 (2023). [2nd author]
8. Tonnah, R. K., Chai, M., Abdollahzadeh, M., Xiao, H., Mohammad, M., et al. Bioinspired Angstrom-Scale Heterogeneous MOF-on-MOF Membrane for Osmotic Energy Harvesting. *ACS Nano* 17(13), 12445–12457 (2023). [7th author]
9. Abdollahzadeh, M., Chai, M., Hosseini, E., **Zakertabrizi, M.**, Mohammad, M., et al. Designing Angstrom-Scale Asymmetric MOF-on-MOF Cavities for High Monovalent Ion Selectivity. *Advanced Materials* 34(9), 2107878 (2022). [4th author]
10. Aramfard, M., Kaynan, O., Hosseini, E., **Zakertabrizi, M.**, Pérez, L. M., Asadi, A. Aqueous Dispersion of Carbon Nanomaterials with Cellulose Nanocrystals: an investigation of molecular interactions. *Small* 18(37), 2202216 (2022). [4th author]
11. Ahmadi, H., **Zakertabrizi, M.**, Hosseini, E., Cha-Umpong, W., Abdollahzadeh, M., et al. Heterogeneous asymmetric passable cavities within graphene oxide nanochannels for highly efficient lithium sieving. *Desalination* 538, 115888 (2022). [2nd author]
12. **Zakertabrizi, M.**, Hosseini, E., Korayem, A. H., Razmjou, A., Fane, A. G., Chen, V. Insight from perfectly selective and ultrafast proton transport through anhydrous asymmetrical graphene oxide membranes under the Grotthuss mechanism. *Journal of Membrane Science* 618, 118735 (2021). [1st author]
13. **Zakertabrizi, M.**, Hosseini, E., Habibnejad Korayem, A., Razmjou, A. Hydrous proton transfer through graphene interlayer: an extraordinary mechanism under magnifier. *Advanced Materials Technologies* 6(10), 2001049 (2021). [1st author]
14. **Zakertabrizi, M.**, Hosseini, E., Sukumaran, S., Korayem, A. H., Fini, E. H. Turning two waste streams into one solution for enhancing sustainability of the built environment. *Resources, Conservation and Recycling* 174, 105778 (2021). [1st author]
15. Kabir, S. F., **Zakertabrizi, M.**, Hosseini, E., Fini, E. H. Effects of amide-based modifiers on surface activation and devulcanization of rubber. *Computational Materials Science* 188, 110175 (2021). [2nd author]
16. Hosseini, E., **Zakertabrizi, M.**, Korayem, A. H., Carbone, P., Esfandiar, A., et al. Mechanical hydrolysis imparts self-destruction of water molecules under steric confinement. *Physical Chemistry Chemical Physics* 23(10), 5999–6008 (2021). [2nd author]
17. Hosseini, E., **Zakertabrizi, M.**, Shariati, S., Rajib, A. I., Fini, E. H. Robust cleaning mechanism permanently detaches hydrocarbon species from silicate surfaces by amphiphiles. *Applied Surface Science* 558, 149954 (2021). [2nd author]
18. Hosseini, E., **Zakertabrizi, M.**, Habibnejad Korayem, A., Zaker, Z., et al. Orbital overlapping through induction bonding overcomes the intrinsic delamination of 3D-printed cementitious binders. *ACS Nano* 14(8), 9466–9477 (2020). [2nd author]

19. Cha-Umpong, W., Hosseini, E., Razmjou, A., **Zakertabrizi, M.**, Korayem, A. H., et al. New molecular understanding of hydrated ion trapping mechanism during thermally-driven desalination by pervaporation using GO membrane. *Journal of Membrane Science* 598, 117687 (2020). [4th author]
20. Hosseini, E., **Zakertabrizi, M.**, Korayem, A. H., Xu, G. A novel method to enhance the interlayer bonding of 3D printing concrete: An experimental and computational investigation. *Cement and Concrete Composites* 99, 112–119 (2019). [2nd author]
21. Hosseini, E., **Zakertabrizi, M.**, Habibnejad Korayem, A., Shahsavari, R. Tunable, multifunctional ceramic composites via intercalation of fused graphene boron nitride nanosheets. *ACS Applied Materials & Interfaces* 11(8), 8635–8644 (2019). [2nd author]
22. Hosseini, E., **Zakertabrizi, M.**, Korayem, A. H., Chen, S., Mohsenabadi, S. K. Graphene oxide in ceramic-based layered structure: Nanosheet optimization. *Construction and Building Materials* 224, 266–275 (2019). [2nd author]
23. Ebrahimi, K., Daiezadeh, M. J., **Zakertabrizi, M.**, Zahmatkesh, F., Korayem, A. H. A review of the impact of micro- and nanoparticles on freeze-thaw durability of hardened concrete: Mechanism perspective. *Construction and Building Materials* 186, 1105–1113 (2018). [3rd author]
24. Hosseini, E., **Zakertabrizi, M.**, Habibnejad Korayem, A., Chang, Z. Mechanical and electromechanical properties of functionalized hexagonal boron nitride nanosheet: A density functional theory study. *The Journal of Chemical Physics* 149(11) (2018). [2nd author]
25. Korayem, A. H., Tourani, N., **Zakertabrizi, M.**, Sabziparvar, A. M., Duan, W. H. A review of dispersion of nanoparticles in cementitious matrices: Nanoparticle geometry perspective. *Construction and Building Materials* 153, 346–357 (2017). [3rd author]

RESEARCH EXPERIENCE

Graduate Research Assistant

Jan 2024 – Present

Department of Mechanical Engineering, Texas A&M University — Advisor: Dr. Matthew Powell-Palm

- **Thermogalvanic cement (Dissertation).** Designed and characterized the first Portland-cement thermogalvanic cell, tailoring pore architecture to balance thermo-electrochemical performance against mechanical integrity.
- **Thermally-driven redox-split mechanism.** Introduced a thermogalvanic mechanism exploiting chemical heterogeneity from localized hot-electrode oxidation stabilized by weakly coordinating anions, delivering 6.44 mV K⁻¹ Seebeck and ~8 W m⁻² peak power sustained over >8 days. Published in *Journal of Materials Chemistry A* and *Nano Letters*.
- **Instrument development.** Designed and fabricated a custom hotwire thermal conductivity measurement prototype for conductive aqueous and gel electrolytes with ±5% accuracy across a wide temperature range.

Graduate Research Assistant

Jan 2022 – Jan 2024

Department of Mechanical Engineering, Texas A&M University — Advisor: Dr. Amir Asadi

- **Nanopatterned Li-ion cathodes.** Designed a spray-deposition system for nanopatterned LiFePO₄ cathodes; introduced a geometric shape factor (ring vs. disk) linking LFP-to-rGO ratio to the structural cohesion–electrochemical performance balance. Published in *ACS Nano*.
- **Multifunctional composites.** Developed molecular-scale friction models for woven nanofabrics; established structure–property relationships linking weave architecture to interfacial friction and epoxy-matrix compatibility. Published in *Advanced Composites and Hybrid Materials*.

Research Assistant

Sep 2015 – Dec 2021

Nanomaterials Research Centre, Iran University of Science and Technology

- **3D-printed concrete.** Co-developed an epoxy-binder to reinforce interlayer bonding by ~84%; used MD simulations to identify Ca²⁺–hydroxyl iono-covalent bonding as the microscale mechanism. Published in *ACS Nano*.
- **Ion-selective membranes.** Co-led an international collaboration with U. of Technology Sydney and Macquarie U. modeling MOF-on-MOF cavities for Li⁺ selectivity, achieving K⁺/Li⁺ and Na⁺/Li⁺ selectivities of 84 and 80. Published in *Advanced Materials*, *Desalination*, and *ACS Nano*.

- **Sustainable bio-binders.** Collaborated with researchers from Arizona State U. to investigate sulfur-radical bonding with bio-oil; DFT/MD modeling explained a 300% increase in complex modulus over control bitumen. Published in *Resources, Conservation and Recycling*.

TEACHING EXPERIENCE

Teaching Assistant — Mechanical Measurements Lab (MEEN 260) Sep 2025 – May 2026

Department of Mechanical Engineering, Texas A&M University

Led laboratory sessions for 40 students; conducted equipment training and safety guidance; graded lab worksheets and rubric-based exams for 200 students.

Teaching Assistant — Construction Materials Lab Jan 2021 – May 2021

Department of Civil Engineering, Iran University of Science and Technology

Adapted laboratory content during the COVID-19 pandemic; developed presentations bridging academic theory and industry-standard material-testing practice.

Teaching Assistant — Structural Loading Aug 2020 – Dec 2021

Department of Civil Engineering, Iran University of Science and Technology

Adapted course content for remote delivery during the COVID-19 pandemic.

Teaching Assistant — Random Vibrations (graduate) Aug 2019 – Dec 2019

Department of Civil Engineering, Sharif University of Technology

Assisted in instruction of a graduate course covering stochastic processes, spectral analysis, and structural dynamics.

TECHNICAL SKILLS

Microscopy & characterization: SEM, EDS (Tescan FERA), FTIR (Bruker ALPHA II), UV-Vis (Genesys 10S), Surface Profiler (Bruker DektakXT); familiar with Zeta Potential Analyzer (Malvern Zetasizer Nano ZS) and AFM (Bruker Dimension Icon).

Electrochemistry: Galvanic & thermogalvanic cell assembly, EIS, Cyclic Voltammetry (CV), Galvanostatic Charge-Discharge (CHI Instruments e600F / 604F).

Computation & modeling: Python (Spyder, SciPy, NumPy, Pandas), MATLAB (image processing, GPIB control), LabVIEW, Abaqus, Materials Studio, Origin Pro; familiar with LAMMPS.

Engineering design: SolidWorks, AutoCAD, ETABS.